Composition-conditioned semantic pairing

Frozen-checkpoint control on composition-eligible test subsets

Training seed 42 · Training seed 7 · Training seed 1234 · Open marker: eligible N < 30 or only one distinct donor map

(a) Exact reduced composition (primary)

J-MBJ · seed 42 (N=167)
J-MBJ · seed 7 (N=167)
J-MBJ · seed 1234 (N=167)
J-OPT · seed 42 (N=679)
J-Formation · seed 42 (N=679)
J-Formation · seed 7 (N=679)
J-Formation · seed 1234 (N=679)
J-Bulk · seed 42 (N=169)
J-Bulk · seed 7 (N=169)
J-Bulk · seed 1234 (N=169)
MP-Bandgap · seed 42 (N=12)
MP-Bandgap · seed 7 (N=12)
MP-Bandgap · seed 1234 (N=12)
MP-Bulk · seed 42 (N=45)

−40 −20 0 20 40

Correct-pairing MAE advantage over conditional pairing (%)

(b) Element set (secondary)

J-MBJ · seed 42 (N=347)
J-MBJ · seed 7 (N=347)
J-MBJ · seed 1234 (N=347)
J-OPT · seed 42 (N=1713)
J-Formation · seed 42 (N=1713)
J-Formation · seed 7 (N=1713)
J-Formation · seed 1234 (N=1713)
J-Bulk · seed 42 (N=370)
J-Bulk · seed 7 (N=370)
J-Bulk · seed 1234 (N=370)
MP-Bandgap · seed 42 (N=90)
MP-Bandgap · seed 7 (N=90)
MP-Bandgap · seed 1234 (N=90)
MP-Bulk · seed 42 (N=117)

−40 −20 0 20 40

Correct-pairing MAE advantage over conditional pairing (%)

Frozen, validation-selected SFTGNN-GPSR checkpoints; each comparison uses its group-eligible test subset. Correct advantage = 100 × (1 − MAE_correct / mean MAE_conditional); positive values favor correct pairing. Error bars are descriptive 95% paired crystal-bootstrap intervals (2,000 resamples; conditional absolute errors averaged within each crystal). Composition-cluster and donor dependence remains. Twenty draws may repeat donor maps. These exploratory within-checkpoint controls do not compare independently trained models. J: JARVIS; MP: Materials Project; MBJ/OPT: band-gap settings; Formation: formation energy; Bulk: bulk modulus.